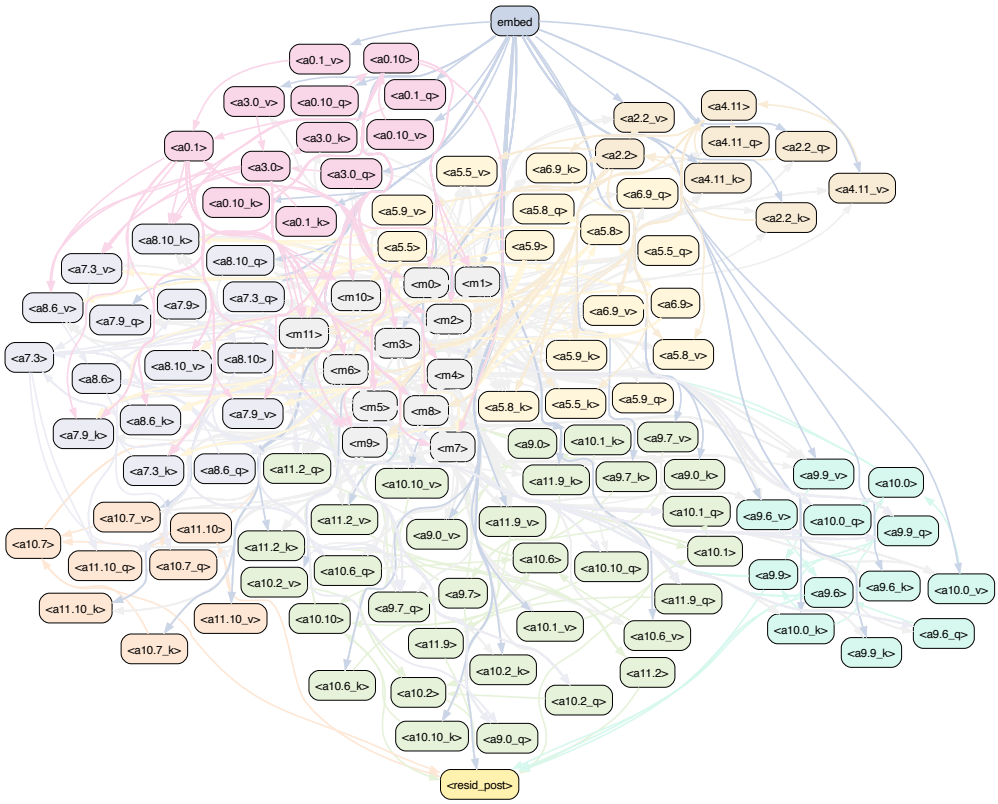


Figure 14: Our low-level implementation of the IOI circuit (Wang et al., 2023), in terms of heads split by query, key, value; and MLPs. It has 1041 edges. For edge between groups A, B in Fig. 13, we connect each member of group A with each member of group B. The colors of groups in this figure correspond to the group colors in Fig. 13.

S-Inhibition Heads and Name Mover Heads) and are sufficient to complete the IOI task, showing that ACDC indeed can recover circuits rather than just subgraphs.

For our ROC plots, we considered the computational graph of IOI described in Figure 13.

The ground-truth circuit gets a logit difference of 3.24 compared to the model's 4.11 logit difference. It has a KL divergence of 0.44 from the original model.

F.3 Limitations of ACDC in recovering the IOI circuit

The main figure from the paper Figure 1 shares several features with circuits recovered with similar thresholds, even when logit difference rather than KL divergence is minimized. The figure does not include heads from all the head classes that Wang et al. (2023) found, as it does not include the Negative Name Mover Heads or the Previous Token Heads. In Figure 15 we run ACDC with a lower threshold and find that it does recover Previous Token Heads and Negative Name Mover Heads, but also many other heads not documented in the IOI paper. This is a case where KL divergence performs better than logit difference maximisation (which does not find Negative Name Movers at any threshold), but still is far from optimal (many extraneous heads are found). Ideally automated circuit discovery algorithms would find negative components even at higher thresholds, and hence we invite future empirical and theoretical work to understand negative components and build interpretability algorithms capable of finding negative components. An early case study gaining wide understanding of a negative head in GPT-2 Small can be found in McDougall et al. (2023).

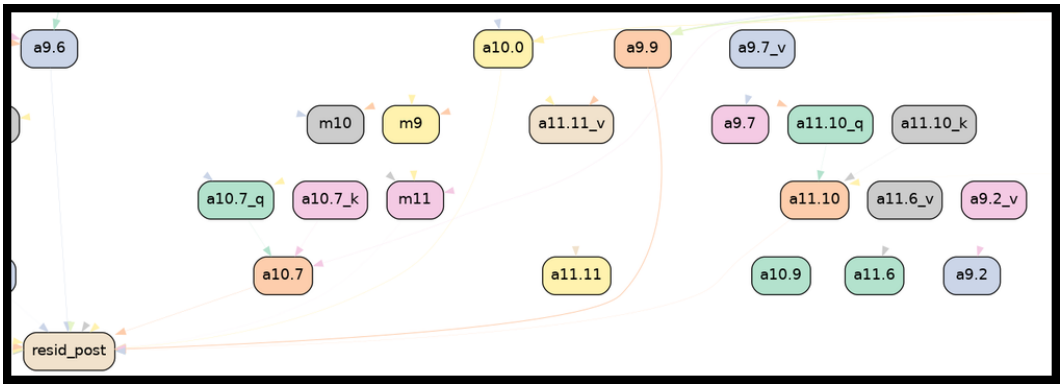
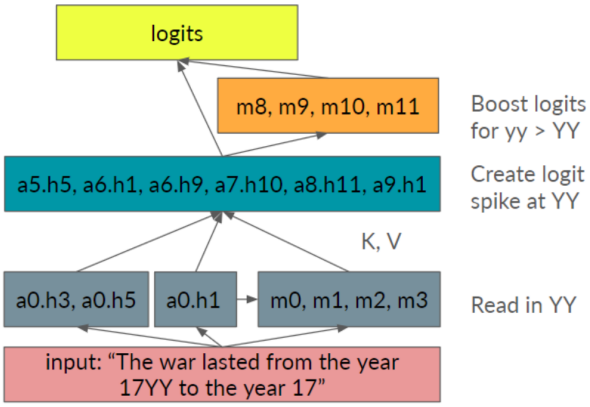
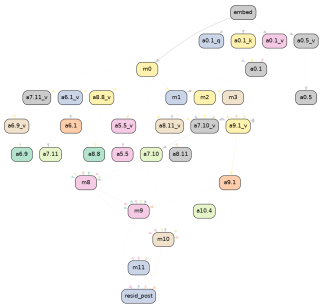


Figure 15: A subset of the 443/32923 edges of GPT-2 Small that ACDC recovered when optimizing for KL divergence at threshold $\tau = 0.00398$. This subset includes edges between Negative Heads (10.7 and 11.10). A number of heads not found by the IOI work (9.2 and 11.11 for example) were also found.



(a) The ground-truth circuit from Hanna, Liu, and Variengien (2023, Figure 13).



(b) Sample Greater-Than graph recovered by ACDC

Figure 16: The ground-truth Greater-Than circuit (16a) and a circuit that ACDC recovers (16b).

G Greater-Than task: details and qualitative evidence

We use a random sample of 100 datapoints from the dataset provided by Hanna, Liu, and Variengien (2023).

We use the circuit from Figure 13 from their paper, including connections between MLPs that are in the same group (e.g MLPs 8, 9, 10 and 11) but not including connections connections between attention heads in the same group. We also include all Q and K and V connections between attention heads present. Their circuit includes all earlier layer connections to the queries of the mid-layer attention heads that cause a logit spike (Figure 16a). This would account for more than half of the edges in the circuit were we to include all such edges that compute these query vectors, and hence we compromised by just adding all early layer MLPs as connections to these query vectors. Full details on the circuit can be found in our codebase[8] and in Fig. 17. This circuit gets a probability difference score of 72% on a subset of their dataset for which GPT-2 Small had an 84% probability difference and a KL divergence from the original model of 0.078.

[8] https://github.com/ArthurConmy/Automatic-Circuit-Discovery/blob/main/acdc/greaterthan/utils.py#L231

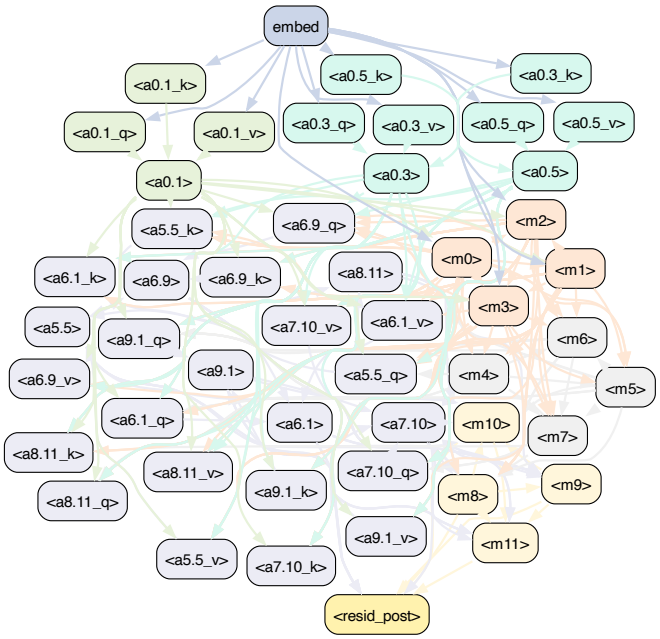


Figure 17: Our low-level implementation of the Greater-Than circuit (Hanna, Liu, and Variengien, 2023), in terms of heads split by query, key, value; and MLPs. It has 262 edges. For edge between groups A, B in Fig. 16a, we connect each member of group A with each member of group B.

An example subgraph ACDC recovered, including a path through MLP 0, mid-layer attention heads and late-layer MLPs, is shown in Figure 16b. This run used the Greater-Than probability difference metric and a threshold of 0.01585 and recovered the results in the abstract.

H Docstring task: details and qualitative evidence

H.1 The docstring circuit

Heimersheim and Janiak (2023) find a circuit in a small language model that is responsible for completing Python docstrings. The model is a 4-layer attention-only transformer model, trained on natural language and Python code. This circuit controls which variable name the model predicts in each docstring line, e.g for the prompt in Table 1 it chooses `shape` over the other variable names `files`, `obj`, `state`, `size`, or `option`.

The circuit is specified on the level of attention heads, consisting of 8 main heads (0.2, 0.4, 0.5, 1.4, 2.0, 3.0, and 3.6) that compose over four layers, although it only makes use of three levels of composition. It consists of 37 edges between inputs, output, and the attention heads, as we show in Figure 18a. We discuss below why we exclude 0.2 and 0.4.

We apply the ACDC algorithm (Section 2) to this dataset, using the prompts from Heimersheim and Janiak (2023) found in their accompanying Colab notebook.[9] For our corrupted dataset we use their `random_random` dataset which randomizes both the variable names in the function definition as well as in the docstring of prompts.

ACDC generates the subgraph shown in Figure 18b. We now compare this to the original 8-head circuit from Heimersheim and Janiak (2023), which was the most specific circuit *evaluated* in that work. We refer to this circuit as the 'manual' circuit to distinguish it from the ground truth, which

[9] Available at https://colab.research.google.com/drive/17CoA1yARaWHvV14zQGcI3ISz1bIRZKS5 as of 8th April 2023